# Multi-Condition Latent Diffusion for Paired Antibody Sequences

*Sampling controls the novelty–foldability trade-off in multi-condition antibody diffusion.*

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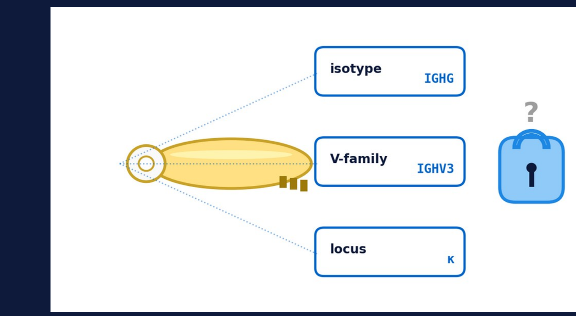


matches paper   our extension

## §1 Why antibody generation is hard

**Need:** valid, diverse, non-memorized paired sequences.

**Design brief** = 3 biological conditions:

- isotype → key handle
- V-gene family → main groove
- light-chain locus → side notch

**Question:** can continuous (LD4LG) and discrete (DPLM) diffusion produce useful variation under the same brief?

## §2 Running example: IGHG · IGHV3 · κ

We trace one antibody-key from brief → generated sequence → folded structure. The same example reappears in §3 and §4.

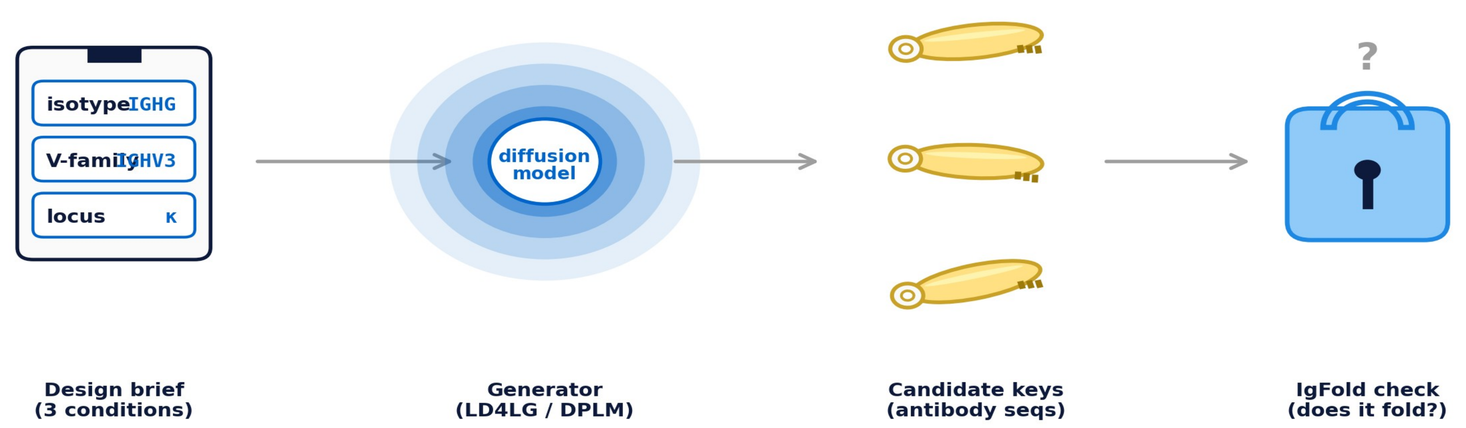


*Fig 1. Design brief → diffusion generator → candidate antibody-keys → IgFold check.*

## §2.1 Why this matters (ML framing)

Token-space Gaussian diffusion (Diffusion-LM) gets MAUVE 0.04 — discrete-noise mismatch. **LD4LG** pushes that boundary into a continuous latent (MAUVE 0.72). **DPLM** instead avoids it via absorbing-state mask noise. We compare both, on the same data.

## §3 Two diffusion paradigms, one task

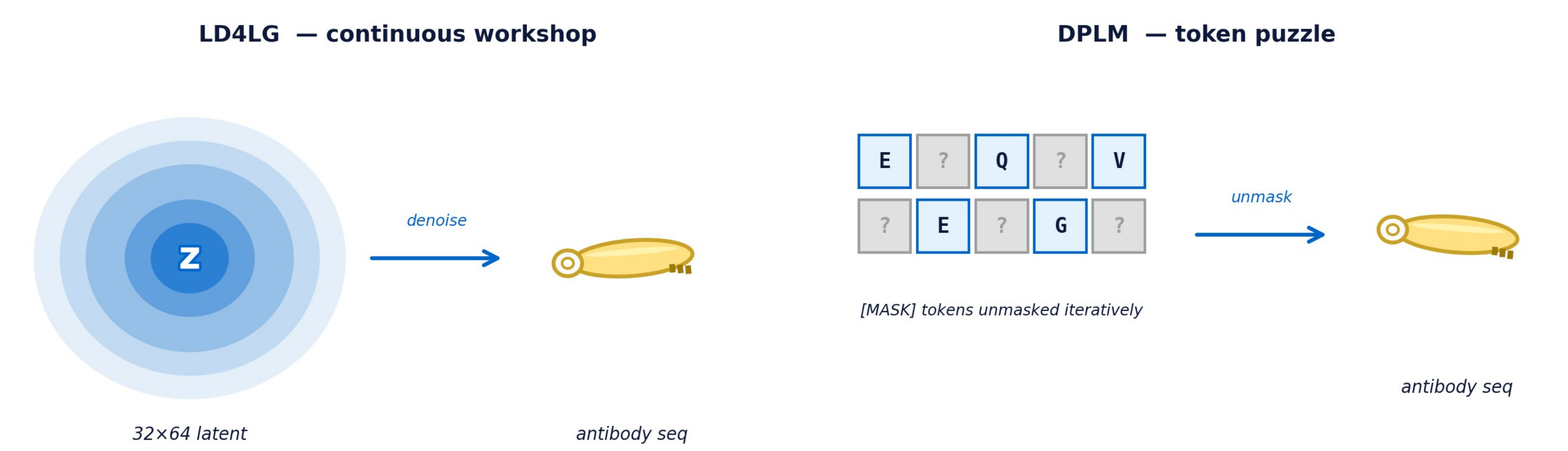


*Fig 2. LD4LG denoises a continuous latent. DPLM unmasks tokens iteratively.*

**LD4LG (continuous workshop)**

- Encoder → 32×64 latent
- Cosine schedule + v-pred loss
- 12-layer denoiser w/ AdaLN(t,c)
- 250-step DDPM
- AR decoder → AA tokens

**DPLM (token puzzle)**

- 1-stage bidirectional Transformer
- Mask schedule γ(t)
- CE loss on masked positions
- 100-step iterative unmask
- **top-p stochastic fix** (our extension)

## §4 The sampling fix (our extension)

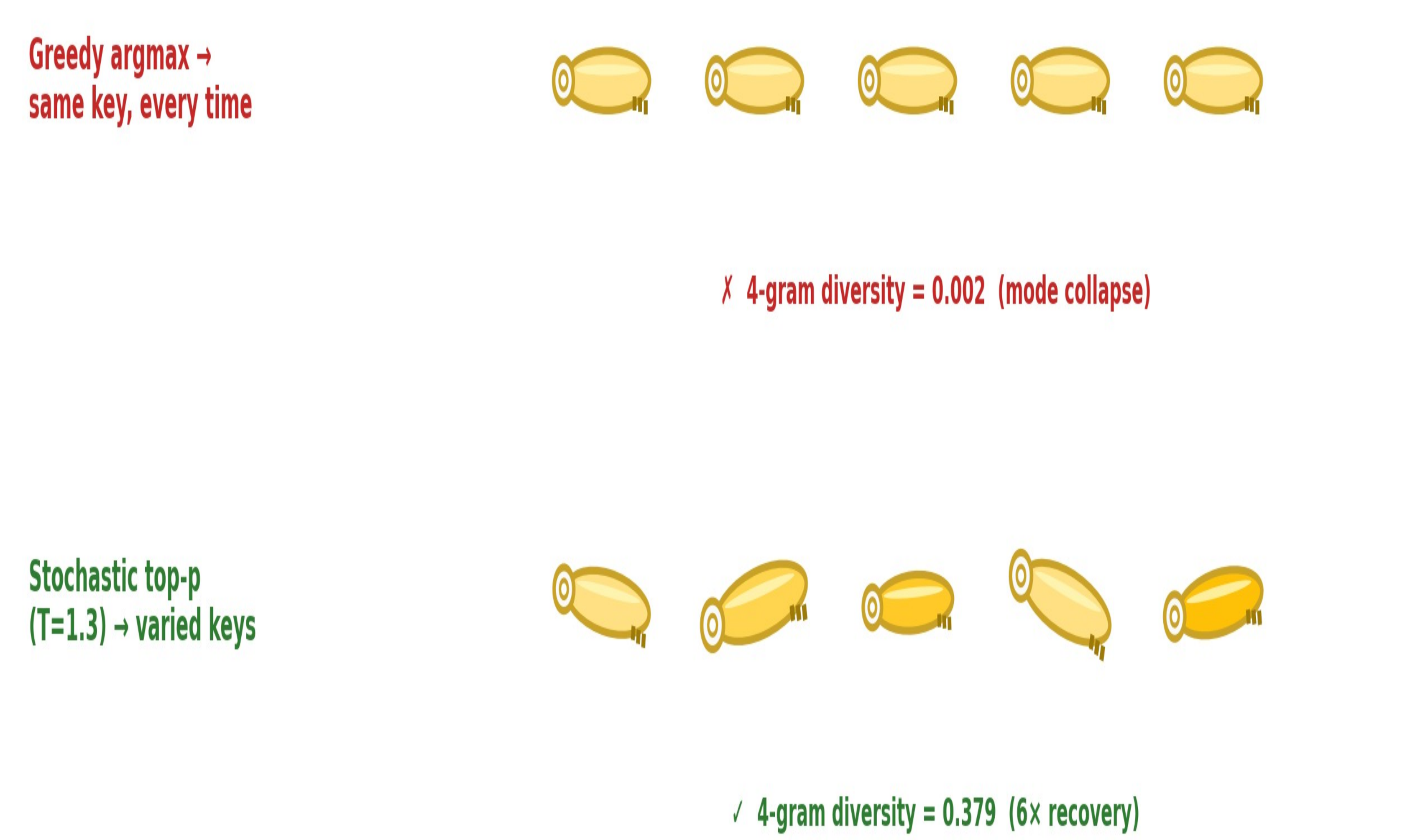


*Fig 3. Greedy keeps minting the same key; top-p sampling unlocks variation. 20-config sweep on top of this.*

## §5 Three findings

**✓ Reproduction works**

**100%** validity · **0%** memorize · **100%** V-gene fid

**✗ Greedy DPLM collapsed**

4-gram diversity = **0.002** *all keys nearly identical*

**Diversity has a cost (our key finding)**

div **0.067 → 0.379** (×6 ↑)

foldable share **97% → 26%** *(intrinsic, not a bug)*

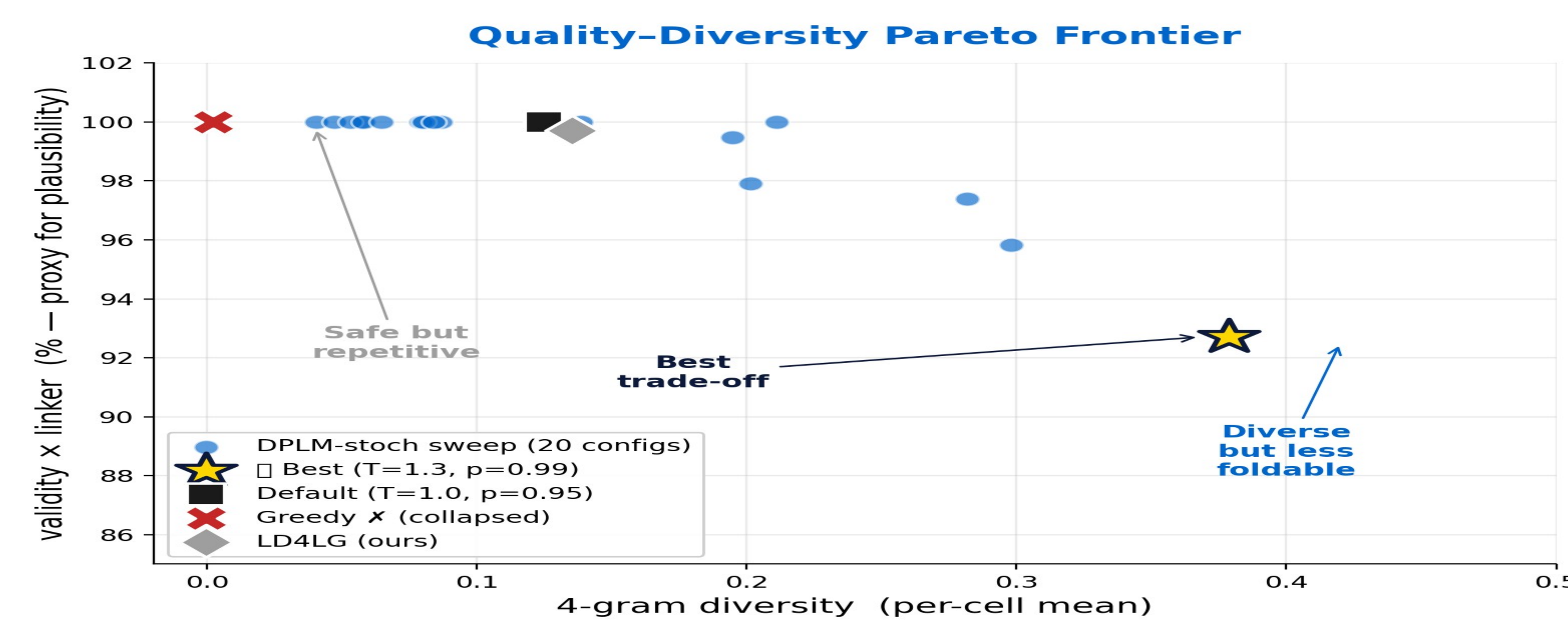


*Fig 4. Pareto frontier — each marker is one sweep config. Novelty starts costing structural plausibility around T=1.15.*

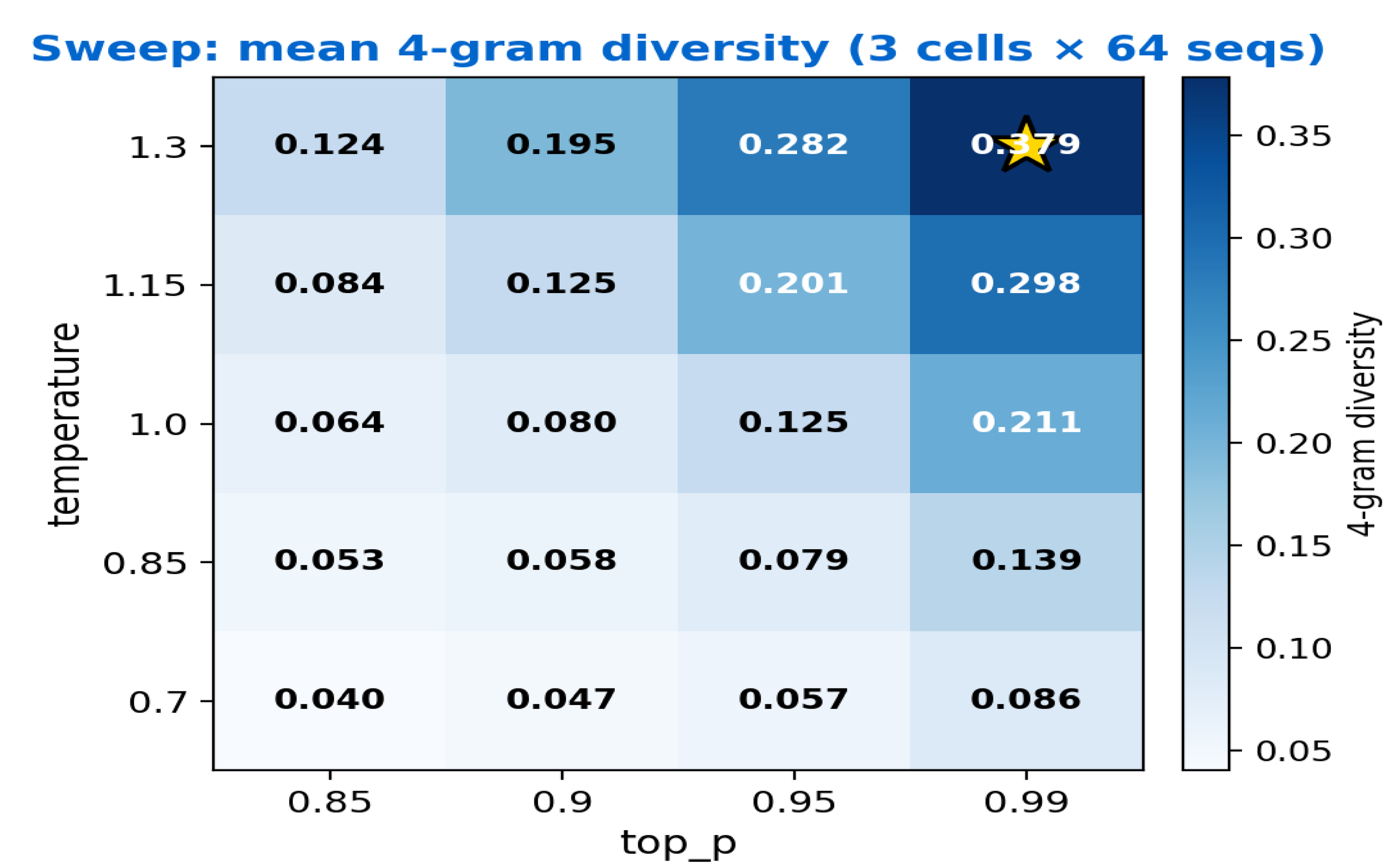


*Fig 5. Sampling is a control knob, not an implementation detail.*

*Diffusion model choice matters, but* ***sampling strategy*** *decides whether the model produces clones, useful variants, or biologically risky sequences.*

Greedy → mode collapse | DDP self-cond → rank divergence | More diversity → less foldability

*Future: antigen-conditional · structure-aware diffusion · active selection on Pareto*

Lovelace et al. LD4LG NeurIPS 2023 · Wang et al. DPLM ICLR 2024 · Ho & Salimans CFG 2022 · Salimans & Ho v-pred ICLR 2022 · Sahoo et al. MDLM 2024 · Ruffolo et al. IgFold Nat. Comms 2023